

1. The cubic equation

$$2x^3 - 3x^2 + 5x + 7 = 0$$

has roots α , β and γ .

Without solving the equation, determine the exact value of

(i) $\alpha^2 + \beta^2 + \gamma^2$ (3)

(ii) $\frac{3}{\alpha} + \frac{3}{\beta} + \frac{3}{\gamma}$ (3)

(iii) $(5 - \alpha)(5 - \beta)(5 - \gamma)$ (3)



9. The cubic equation

$$3x^3 + x^2 - 4x + 1 = 0$$

has roots α , β , and γ .

Without solving the cubic equation,

(a) determine the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ (3)

(b) find a cubic equation that has roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$, giving your answer in the form

$x^3 + ax^2 + bx + c = 0$, where a , b and c are integers to be determined. (3)



4. The cubic equation

$$x^3 + 3x^2 - 8x + 6 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(\alpha - 1)$, $(\beta - 1)$ and $(\gamma - 1)$, giving your answer in the form $w^3 + pw^2 + qw + r = 0$, where p , q and r are integers to be found.

(5)

2. The cubic equation

$$z^3 - 3z^2 + z + 5 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(2\alpha + 1)$, $(2\beta + 1)$ and $(2\gamma + 1)$, giving your answer in the form $w^3 + pw^2 + qw + r = 0$, where p , q and r are integers to be found.

(5)

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2. The cubic equation

$$2x^3 + 6x^2 - 3x + 12 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(\alpha + 3)$, $(\beta + 3)$ and $(\gamma + 3)$, giving your answer in the form $pw^3 + qw^2 + rw + s = 0$, where p , q , r and s are integers to be found.

(5)



10.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) The quartic equation

$$z^4 + 5z^2 - 30 = 0$$

has roots p, q, r and s .

Without solving the equation, determine the quartic equation whose roots are

$$(3p - 1), (3q - 1), (3r - 1) \text{ and } (3s - 1)$$

Give your answer in the form $w^4 + aw^3 + bw^2 + cw + d = 0$, where a, b, c and d are integers to be found.

(5)

(ii) The roots of the cubic equation

$$4x^3 + nx + 81 = 0 \quad \text{where } n \text{ is a real constant}$$

are $\alpha, 2\alpha$ and $\alpha - \beta$

Determine

(a) the values of the roots of the equation,

(5)

(b) the value of n .

(2)



4. The roots of the quartic equation

$$3x^4 + 5x^3 - 7x + 6 = 0$$

are α , β , γ and δ

Making your method clear and without solving the equation, determine the exact value of

(i) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ (3)

(ii) $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} + \frac{2}{\delta}$ (3)

(iii) $(3 - \alpha)(3 - \beta)(3 - \gamma)(3 - \delta)$ (3)



Answer ALL questions. Write your answers in the spaces provided.

1. $f(z) = z^3 + pz^2 + qz - 15$

where p and q are real constants.

Given that the equation $f(z) = 0$ has roots

$$\alpha, \frac{5}{\alpha} \text{ and } \left(\alpha + \frac{5}{\alpha} - 1 \right)$$

(a) solve completely the equation $f(z) = 0$

(5)

(b) Hence find the value of p .

(2)

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7. $f(z) = z^3 - 8z^2 + pz - 24$

where p is a real constant.

Given that the equation $f(z) = 0$ has distinct roots

$$\alpha, \beta \text{ and } \left(\alpha + \frac{12}{\alpha} - \beta \right)$$

(a) solve completely the equation $f(z) = 0$ (6)

(b) Hence find the value of p . (2)



7.

$$f(z) = z^3 + z^2 + pz + q$$

where p and q are real constants.

The equation $f(z) = 0$ has roots z_1, z_2 and z_3

When plotted on an Argand diagram, the points representing z_1, z_2 and z_3 form the vertices of a triangle of area 35

Given that $z_1 = 3$, find the values of p and q .

(7)

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6. An art display consists of an arrangement of n marbles.

When arranged in ascending order of mass, the mass of the first marble is 10 grams. The mass of each subsequent marble is 3 grams more than the mass of the previous one, so that the r th marble has mass $(7 + 3r)$ grams.

- (a) Show that the mean mass, in grams, of the marbles in the display is given by

$$\frac{1}{2}(3n+17) \quad (3)$$

Given that there are 85 marbles in the display,

- (b) use the standard summation formulae to find the standard deviation of the mass of the marbles in the display, giving your answer, in grams, to one decimal place.

(6)



6. (a) Use the standard results for $\sum_{r=1}^n r^2$ and $\sum_{r=1}^n r$ to show that

$$\sum_{r=1}^n (3r-2)^2 = \frac{1}{2}n[6n^2 - 3n - 1]$$

for all positive integers n .

(5)

(b) Hence find any values of n for which

$$\sum_{r=5}^n (3r-2)^2 + 103 \sum_{r=1}^{28} r \cos\left(\frac{r\pi}{2}\right) = 3n^3$$

(5)

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5.

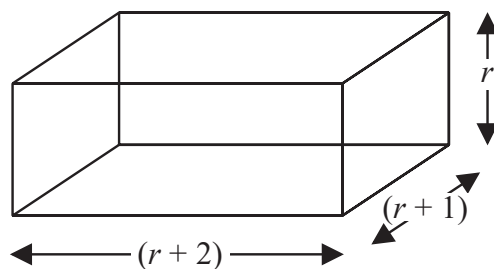


Figure 2

A block has length $(r + 2)$ cm, width $(r + 1)$ cm and height r cm, as shown in Figure 2.

In a set of n such blocks, the first block has a height of 1 cm, the second block has a height of 2 cm, the third block has a height of 3 cm and so on.

- (a) Use the standard results for $\sum_{r=1}^n r^3$, $\sum_{r=1}^n r^2$ and $\sum_{r=1}^n r$ to show that the **total** volume, V , of all n blocks in the set is given by

$$V = \frac{n}{4}(n+1)(n+2)(n+3) \quad n \geq 1 \quad (5)$$

Given that the total volume of all n blocks is

$$(n^4 + 6n^3 - 11\,710) \text{ cm}^3$$

- (b) determine how many blocks make up the set. (2)



9. The cubic equation

$$3x^3 + x^2 - 4x + 1 = 0$$

has roots α , β , and γ .

Without solving the cubic equation,

(a) determine the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ (3)

(b) find a cubic equation that has roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$, giving your answer in the form

$x^3 + ax^2 + bx + c = 0$, where a , b and c are integers to be determined. (3)



Answer ALL questions. Write your answers in the spaces provided.

1. $f(z) = z^3 + pz^2 + qz - 15$

where p and q are real constants.

Given that the equation $f(z) = 0$ has roots

$$\alpha, \frac{5}{\alpha} \text{ and } \left(\alpha + \frac{5}{\alpha} - 1\right)$$

(a) solve completely the equation $f(z) = 0$

(5)

(b) Hence find the value of p .

(2)

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5. (a) Use the standard summation formulae to show that, for $n \in \mathbb{N}$,

$$\sum_{r=1}^n (3r^2 - 17r - 25) = n(n^2 - An - B)$$

where A and B are integers to be determined.

(4)

- (b) Explain why, for $k \in \mathbb{N}$,

$$\sum_{r=1}^{3k} r \tan(60r)^\circ = -k\sqrt{3}$$

(2)

Using the results from part (a) and part (b) and showing all your working,

- (c) determine any value of n that satisfies

$$\sum_{r=5}^n (3r^2 - 17r - 25) = 15 \left[\sum_{r=6}^{3n} r \tan(60r)^\circ \right]^2$$

(6)



